Revolutionizing Research Writing and Publishing by using AI-Powered Tools and Techniques

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Abstract:

The main purpose of this article is to analysis the functions of AI-driven tools which are useful to the researcher for their research communications, its challenges, benefits as well as applications in the era of digital transformation. This analysis started from Literature review in the field of Artificial Intelligence related with digital switch-over of various sector specially related to education sector, to find out the key tools, their functions, features and how does researchers can accompanied their work for further research with AI-driven tools.

The paper covers the role of artificial intelligence (AI) in scholarly publishing. It investigates diverse artificial intelligence techniques and their applications in tasks such as literature evaluation, article preparation, plagiarism prevention, peer review, and publishing production. The material emphasizes the relevance of human-machine collaboration in the publishing industry, as well as the potential of Artificial Intelligence (AI) to improve the nature and productivity of research communication. These AI-based research utility tools have the best attributes for saving researchers time and effort, as well as for facilitating technical communication. However, they also have equal potential drawbacks, which are detrimental to science and technology.

Keywords: Artificial intelligence, Artificial Intelligence Tools, Research communication, Scholarly publication

1. Introduction:

Majority of scientist are realize and rather than confident about the potentiality of artificial intelligence (AI) as a useful tool fordata analysis and literature reviews. In recent study it was found that for systematic literature reviews (SLR), use of artificial intelligence tools are increasing**(Burger et al., 2023)**. Research in academia can be methodically and effectively supported by AI. Even though technology is still in its formative stages, AI is already demonstrating significant assurance and has the

potentiality to completely alter how we conduct research, especially for no-code applications **(Calo, 2017).**Growth of Artificial Intelligence in recent era give us hope to think that technology at one point can be used to advance and expand research without the need for any technical expertise **(Dwivedi et al., 2023).**

1.1 What Is AI?

John McCarthy first coined the term artificial intelligence in 1956. The English Oxford Living Dictionary gives this definition: "The theory and development of computer systems able to perform tasks normally requiring human intelligence, such as visual perception, speech recognition, decision-making, and translation between languages." Artificial intelligence (AI) is the technology based on a computer or a robot controlled by mechanism to do different intelligent works that are usually done by human beings as it require human intelligence and discernment, though its impossible to do variety of tasks by technology which can ordinarily done by the human beings**(Marr, 2018)**

1.2 What is Machine Learning?

In the last decade, Machine Learning is one of the most popular artificial intelligence technology which is mostly used by technical community. Main function of Machine Learning (ML) is that, it gathers common algorithms from awhole family of algorithms to learn by themselves by observing data. These algorithms areinspired by different sciences and especially by statistics**(Yassine et al., 2023)**

Machine Learning working as a knowledge discovery tools to find out knowledge and make smart decisions. Many practical applications depend on it, including autonomous control systems, computer science and data extraction, recognition systems, and recommendation systems (Bishop, 2006).

1.3 What is an AI Tools?

The definition of AI tools and what they do itself are still evolving, especially with the integration of numerous cutting-edge technologies and data, such as big data and the Internet of Things (Wang et al. 2019a, b). A growing body of research is being done on various facets of AI tools, with a particular emphasis on design of AI tools in a variety of contexts, including supply chain (Priore et al. 2019), biomedicine **(Kocheturov et al. 2019),** and smart healthcare in clinical settings (Pan et al. 2019). These contexts range from requirements elicitation to technical aspects. It's interesting to see how these AI-based tools perform, especially in comparison to earlier methods and algorithms **(Razzaghi et al. 2019).**

According to Synthesia glossary "an AI tool is a software application that uses artificial intelligence algorithms to perform specific tasks and solve problems.AI tools can be used in a variety of

industries, from healthcare and finance to marketing and education, to automate tasks, analyze data, and improve decision-making".

2. AI integration into research workflow

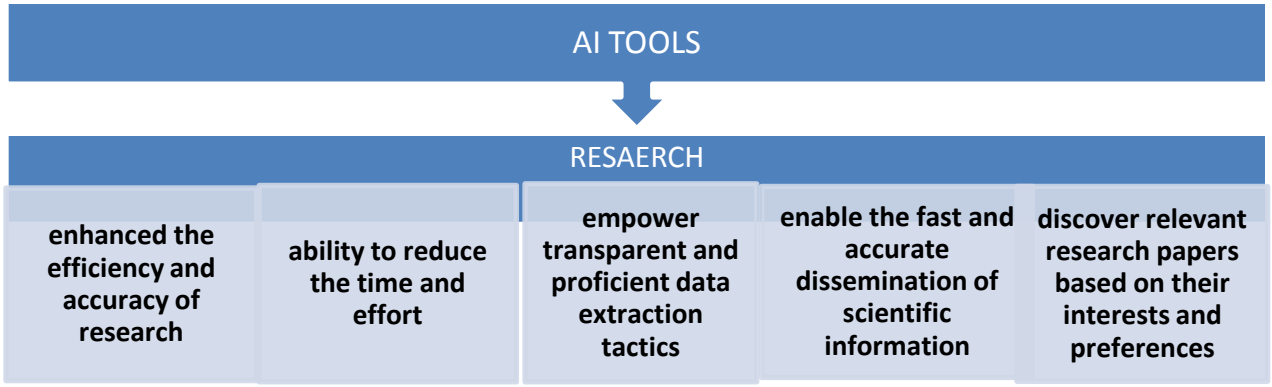

Figure 1. Benefits of using AI Tools in Research

2.1 Why integrate AI in the research process?

AI can be used to detect plagiarism in scientific articles by utilizing tools and algorithms that can identify content similarities at different levels. These AI-powered solutions are superior to traditional plagiarism detection methods and regular web crawlers. Some AI systems can analyze bar chart images, locate paraphrased text, and identify plagiarism in many languages.

Artificial intelligence (AI)-based Intelligent Personal Assistants (IPAs) have many characteristics that make life easier for researchers and appear to be fantastic helpers. More crucially, by utilizing Natural Language Processing (NLP), these virtual assistants respond to user inquiries through voice commands that resemble real-world conversations. The use of natural language processing in these systems gives researchers the impression that they are conversing with actual people.

Researchers have the chance to communicate with a machine that acts like a human through the application of NLP. This may appear to be a substantial advancement in human-machine interaction because these assistants have a big part to play in interpreting the users' vocal commands, defining them, transmitting them, and locating the appropriate information in the form of a semantic query and answer. However, there are still significant challenges to communication and coordination between the user and the system, which makes mutual understanding difficult to accomplish. **(Kondratyuk, 2017).**

3. Some prominent fields where researchers can use AI-driven tools in the modern era:

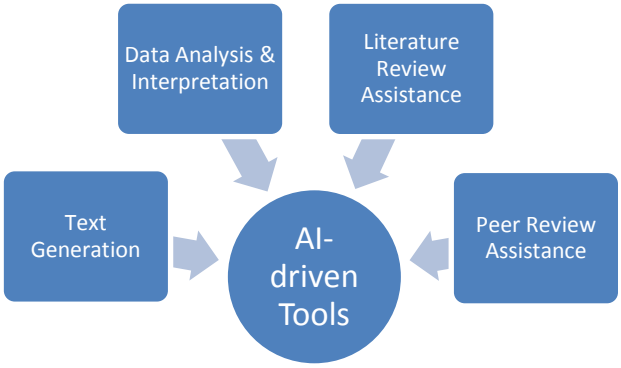

Figure 2 some prominent area of using AI tools

3.1 Assisting in Academic Writing:

Developments of AI technology have revolutionized the process of scientific writing. These AI models can create high-quality research publications if trained with large volumes of scientific data. Researchers may upload data such as research papers and journal articles to produce scientific data. The data is then analyzed and synthesized by the AI model to produce a coherent, pertinent scientific paper that faithfully reproduces the input data. As they no longer need to pore through study papers or create specific parts, this can save investigators a significant amount of time and work. Additionally, since AI-generated scientific writings can use complex NLP techniques and are trained using a significant quantity of scientific data, they are typically of a high caliber **(Solangi et al., 2018), (Paranjape et al., 2019)**

If adequate data and references are available, the derived text is probably to be accurate and relevant, making it a great tool for researchers who want to create scientific articles. Depending on the circumstance, it may also be taught to write in a certain tone, such as formal or informal, or to be concise or descriptive. AI has the potential to raise the standard of publications. It also helps researchers to make sure that their abstracts, introductions, and conclusions are factual and pertinent.

Cost-effectiveness is another advantage of AI-generated scientific texts. A scientific publication might take a lot of time and money to prepare, because researcher has to pay high remunerations to the editors or writers. Researchers can cut costs by using AI-generated tools.By utilizing these tools, Scholars can decrease costs by doing their own writing and editing instead of hiring outside experts.

These technologies, in general, also have limitations.Models may find it challenging to comprehend the nuances and complexity of scientific topics, and as a result, their output may be less sophisticated and unique than literature produced by humans. It is imperative to address moral difficulties, such as unintended plagiarism or creating "hallucinated" knowledge**(Alqahtani et al., 2023)(Crothers et al., 2022).**

List of few AI-driven tools which are useful for generating text:

Name of the APP	**Model**	**Description**	**Website**	**Free/Paid**
Jasper	GPT	One of the most well-known names in AI text generation is Jasper. It excels in producing top-notch material of any length that can be altered to best reflect of brand voice.	https://www.jasper.ai/	Paid
Writesonic	GPT	Writesonic is an all-in-one content creation platform offering text, audio, and image generation.	https://writesonic.com/	10,000 word per month free
Copy.ai	GPT	Another excellent online AI text generator is Copy.ai. Because it offers more than 90 templates, one can create copy for any purpose. Its specifically question-based methodology is fantastic for producing high-quality material quickly.	https://www.copy.ai/	Free 2000 word per month
Quillbot	Paraphrasing/Summerizing	Quillbot is an AI writing tool that enables users to create excellent material quickly. It creates personalised content automatically using an effective algorithm.	https://quillbot.com/	Free
ChatGPT	GPT	ChatGPT can respond to specific inquiries on the subject of your study. It can also help in the early phases of a literature review by recommending important papers, authors, or trends.	https://chat.openai.com/	Free(GPT 3.5)
BARD	GPT	Google introduced Bard in February 2023 as an experimental conversational AI service powered by LaMDA (Language Models for Dialogue Applications).	https://bard.google.com/	Free
HiveMind	GPT	HiveMind can assist you manage projects and develop strategic initiatives in addition to writing articles and responding to emails.	https://hive.com/hivemind/	Free for Individual use

		The newest addition to Hive's platform for workplace efficiency, HiveMind, enables teams to fully utilise AI technology to speed up workflow, make workdays go more smoothly, and create content.		

3.2 Assisting in Data Analysis and Interpretation:

Artificial Intelligence and Natural Language Processing (NLP) are increasingly being used in clinical and scientific settings as a way to explore huge amounts of facts and identify patterns that might otherwise remain undiscovered. In the healthcare sector, for instance, NLP may extract crucial information from unstructured clinical records, such as the reasons why some treatments fail or specific medical conditions.

Although AI and NLP technologies have a lot of potential to improve healthcare outcomes, there are certain significant drawbacks and difficulties that need to be resolved before they can be widely used.

Transformer-based chatbots must be improved for clinical application, and privacy issues involving patient data must be taken into account(**Yeung et al., 2023**).

List of few AI-driven tools which are useful for data analysis and interpretation:

Name of the APP	**Model**	**Description**	**Website**	**Free/Paid**
RapidMiner	Data Science and Machine Learning	For users of all skill levels, data analysis is made simpler by Rapidminer's user-friendly interface and drag-and-drop framework creator. It aids data teams all along the analytics cycle. Access, loading, and analysis of data may be made easier for enterprises because to its integration and machine learning features. RapidMiner allows you to analyze any kind of data, including text, images, and audio recordings.	https://rapidminer.com/	Free with limited features
Tableau	Data Visualization	Tableau makes studying and presenting data visually simple by offering dynamic dashboards and intuitive data visualization. Users can ask	https://www.tableau.com/	Tableau public edition is free

		questions using Ask Data, an AI-powered tool, and receive prompt, visual responses. Users may interact with their data without being to know how to code.		
Google Data Studio	Data Visualization	With this tool one can make the line chart, bar chart, area chart, and pie chart which is free and simple to use. You can also produce as many reports as you like.	https://www.google.com/analytics/data-studio	Free
Polymer	NLP	With only a few clicks, Polymer's powerful AI can evaluate the data and help in better comprehend it. Simply submit spreadsheet to Polymer, and it will be converted into a simplified database that is simple to navigate.	https://app.polymersearch.com/	Commercial
KNIME	Data Science and Machine Learning	An open-source platform offers a complete toolkit for evaluating data, including records preparation, device familiarisation, and visualisation. Users may customise their experience and have flexibility due to integration with different machine learning algorithms and libraries. Collaboration tools and workflow automation support productive analyst teamwork and knowledge exchange.	https://www.knime.com/knime-analytics-platform	Free
Microsoft Power BI	Data Science and Machine Learning	It enables customers to go through their data and visualise it for insights, making it a very effective business intelligence platform. Users of the platform can quickly start creating reports and dashboards after importing data from almost any source.	https://powerbi.microsoft.com/en-us/	Free

3.3 Applying AI to Expedite Literature Reviews:

Every research project must go through the process of analyzing the body of knowledge since it enables the discovery of studies that are relevant to a certain research question. However, the cost and time involved with this process can be high.Development of AI makes it quicker and more effective**(Haman et al., 2023**). The main benefit of AI for analyzing scientific literature is its capacity for processing massive amounts of data quickly and effectively, which saves researchers' time to work on other elements of their studies and saves them time. NLP techniques for keyword and concept identification enable AI systems to precisely locate pertinent studies, uncovering literature that human reviewers could pass over. Additionally, by evaluating study design, sample size, and statistical techniques, AI systems may evaluate the caliber of scientific literature, helping researchers in deciding a study's dependability and its applicability to their review. They can also spot trends and patterns, pointing out areas for future research or knowledge gaps. **(Hosseini et al., 2023).**

Despite these advantages, there are certain disadvantages to using AI and NLP in the literature review process.AI systems may not fully comprehend the context and nuance contained in the scientific literature, which could lead to the inclusion of irrelevant research or the omission of significant ones. The caliber and quantity of training data has an impact on AI system performance as well. The artificial intelligence system may not accurately depict the literary landscape if the training data is incorrect or insufficient.

Furthermore, AI-generated reviews of the literature might lack the knowledge and critical thinking that human reviewers bring to the table, both of which are essential for synchronizing complex material and identifying pertinent links between studies**(Korteling, et al. 2021).**

List of few AI-driven tools which are useful for Literature review:

Name of the APP	**Model**	**Description**	**Website**	**Free/Paid**
Research Rabbit	GPT	Research Rabbit is an AI-powered tool which assists researchers in finding, organising, and analysing research publications. It uses AI to search the web for scholarly publications that are pertinent to its users. Additionally, it helps to save and arrange items in your own personal library.	https://www.researchrabbit.ai/	Free
Elicit	GPT	Elicit was created by Ought, a non-profit machine learning (ML) research lab with headquarters in the US. Elicit collects articles from Semantic Scholar and speeds up	https://elicit.com/	Free

		the literature review process. It is great for evidence synthesis and text extraction.		
Scite	GPT	Scite is a free, open-source AI-powered tool that aids in the discovery and comprehension of research articles. Scite does this by offering Smart Citations, which provide the context of a citation and indicate whether the article contains either favourable or adverse evidence.	https://scite.ai/	Commercial
Scholarcy	GPT	Scholarcy is an AI-powered app that can help you with your academic reading. It can automatically summarize articles, create flashcards, and generate bibliographies. Scholarcy can also help you find related research and identify key information in articles.	https://www.scholarcy.com/	Free
KNIME	Data Science and Machine Learning	An open-source platform offers a complete toolkit for evaluating data, including records preparation, device familiarisation, and visualisation. Users may customise their experience and have flexibility due to integration with different machine learning algorithms and libraries. Collaboration tools and workflow automation support productive analyst teamwork and knowledge exchange.	https://www.knime.com/knime-analytics-platform	Free
Microsoft Power BI	Data Science and Machine Learning	It enables customers to go through their data and visualise it for insights, making it a very effective business intelligence platform. Users of the platform can quickly start creating reports and dashboards after importing data from almost any source.	https://powerbi.microsoft.com/en-us/	Free

3.4 Artificial Intelligence to Improve Peer Review:

NLP and AI can greatly enhance the peer review process for scientific articles, resolving a number of problems and enhancing the overall standard of published research. By employing AI's capabilities, the peer review process can be rendered impartial, effective, accurate, and objective.By automating processes like plagiarism detection and data fabrication identification, AI can expedite the review process while preserving the integrity of the research. Another benefit is the potential rise in peer reviewquality.The research topic, technique, and statistical analysis can all be thoroughly and objectively analyzed by AI systems. Authors can acquire accurate assessments and illuminating critiques by taking a thorough approach, which will help them improve their writing**(Hosseini et al., 2023).**

Journal editors can benefit from chatbots driven by AI in several ways, such as ChatGPT. Taking care of routine duties, including responding to preliminary inquiries, gathering submissions, and reminding authors to submit their work, frees editors to focus on improving the quality of the content, developing relationships with authors, and coming up with editorial strategies. Additionally, chatbots can help editors produce or update content and find manuscript issues quickly by assisting in language editing and proofreading. Finally, chatbots can help authors and reviewers by customizing the publication process by looking at their submission history and offering suggestions on how to prepare and evaluate manuscripts, improve the quality of submissions, and enhance reviewer comments.

AI has also been applied to scholarly publishing to enhance the process of evaluating quality. AI-powered peer review systems, like Frontiers' AIRA, have been created to make suggestions for grammar, style, figure, and legend improvements and to look for possible instances of plagiarism. Platforms like PubSURE Report and ripe-taReview assess reporting standards, characteristics of repeatability, and other elements.Overall, the effectiveness, productivity, and efficiency of academic publishing processes have increased due to AI and AI-powered technologies, making it simpler for researchers to gather crucial information, write papers, identify plagiarism, and ensure high-quality publications.

Application of AI in Peer review process has also some drawbacks,may provide unique difficulties. The lack of intricacy or context particular to various research topics in these systems is one potential disadvantage that could result in misconceptions or omissions. Additionally, biased algorithms and the automation of a process that is typically labor-intensive and centred on human interaction are two ethical conundrums that AI may create. Last but not least, using AI in peer review may encounter opposition from researchers and reviewers who are concerned of replacing human beings.

Furthermore, because LLMs are trained on historical data, they are inherently conservative; hence systems utilizing such models run the risk of reproducing or amplifying preexisting biases. Concerns regarding data security, copyright, intellectual property rights, and the privacy of writers and research subjects are brought up by the lack of transparency surrounding how models use data and prompts. And while the technology's quick advancement is desired or perhaps necessary to stay up with research frontiers, it also implies that AI-assisted reviewing cannot always be replicable.

4. Conclusions

The academic community of today has been greatly impacted by AI tools. Numerous aspects of scientific publication have changed as a result of these tools, including article preparation, plagiarism detection, and quality evaluation.

GenAI tools are anticipated to revolutionize how research is conducted and assessed as a result of their widespread use and ongoing progress. It's critical to keep an eye on and do research on the tools themselves as we proceed in this path.

The unexpected biases that are hard-coded into these models must be understood by researchers. More research is required to increase transparency on prejudices like sexism, racism, and ageism. This could have an effect on like literature reviews and other tasks that are generated by computers.

Future research into the development of the right prompting abilities is another intriguing subject. It appears possible that alternative prompting techniques could exist, resulting in various outputs and possibly various findings from the human researcher.

Finally, GenAI may be a valuable tool for accelerating research reporting. Senior academics are aware of how important and time-consuming each paper's writing is,creating good marketing papers for prestigious journals is frequently compared to a craft that takes time to perfect and is difficult to scale

References

Ahmed, I., Kajol, M., Hasan, U., &Datta, P. P. (2023). ChatGPT vs. bard: A Comparative Study. TechRxiv [Preprint]. https://doi.org/10.36227/techrxiv.23536290.v1

Burger, B., Kanbach, D. K., Kraus, S., Breier, M., &Corvello, V. (2023). On the use of AI-based tools like ChatGPT to support management research. European Journal of Innovation Management, *26*(7), 233–241.https://doi.org/10.1108/EJIM-02-2023-0156

Calo, Ryan, Artificial Intelligence Policy: A Primer and Roadmap (2017). Available at: https://ssrn.com/abstract=3015350 or http://dx.doi.org/10.2139/ssrn.3015350

Christopher M. Bishop. Pattern Recognition and Machine Learning. Springer, New York. 1-758p.

Crothers, E. N., Japkowicz, N., & Viktor, H. L(2022). Machine generated text: A comprehensive survey of threat models and detection methods. 11, 70977–71002. https://doi.org/10.48550/ARXIV.2210.07321

Chubb, J., Cowling, P., & Reed, D. (2022). Speeding up to keep up: Exploring the use of AI in the research process. AI and Society, 37(4), 1439–1457. https://doi.org/10.1007/s00146-021-01259-0

Copeland B.J. (2020). Artificial intelligence. Encyclopaedia Britannica [Internet]. [cited 2023 Aug 8]. Available from: https://www.britannica.com/technology/artificial-intelligence)

Dwivedi, Y. K., Kshetri, N., Hughes, L., Slade, E. L., Jeyaraj, A., Kar, A. K., Baabdullah, A. M., Koohang, A., Raghavan, V., Ahuja, M., Albanna, H., Albashrawi, M. A., Al-Busaidi, A. S., Balakrishnan, J., Barlette, Y., Basu, S., Bose, I., Brooks, L., . . . Wright, R. (2023). Opinion Paper: "So what if ChatGPT wrote it?" Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. International Journal of Information Management, 71, 102642. https://doi.org/10.1016/j.ijinfomgt.2023.102642

Gołąb-Andrzejak, E. (2023). AI-powered digital transformation: Tools, benefits and challenges for marketers – Case study of LPP. Procedia Computer Science, 219, 397–404. https://doi.org/10.1016/j.procs.2023.01.305

Haman, M., &Skolník, M. (2023). doi:10.1080/08989621.2023.2185514. Using ChatGPT to conduct a literature review. Accountability in Research, 1–3. https://doi.org/10.1080/0898962120232185514

Hosseini M.&Horbach S.P.J.M.(2023). Fighting reviewer fatigue or amplifying bias?Considerations and recommendations for use of ChatGPT and other Large LanguageModels in scholarly peer review.https://doi.org/10.21203/RS.3.RS-2587766/V1.

Hosseini, M., Rasmussen, L. M., &Resnik, D. B.. (2023). Using AI to write scholarly publications. Accountability in Research, 1–9. https://doi.org/10.1080/0898962120232168535.

Kocheturov,A., Panos M., Pardalos, P.M. & Karakitsiou, A.(2019). Massive datasets and machine learning for computational biomedicine: trends and challenges. Annals of Operations Research. **276**, 5–34

Korteling JE (Hans), van de Boer-Visschedijk, G.C.,Blankendaal R.A.M., Boonekamp R.C.&Eikelboom, A.R.(2021). Human- versus artificial intelligence. Front Artif Intell.4. https://doi.org/10.3389/FRAI.2021.622364

Lambrecht, A., & Tucker, C. (2019). Algorithmic bias? An empirical study of apparent gender-based discrimination in the display of STEM career ads. *Management Science*, *65*(7), 2966–2981.

Lovrinovic, C. &Volland, H. (2019). The Future impact of artificial intelligence on the publishing industry. Hamburg: Gould Finch and Frankfurter Buchmesse, 24p.

Marr, B.(2018). The key definitions of artificial intelligence (AI) that explain its importance. Forbes [Internet]. [cited 2023 Aug 8]. Available from: https://www.forbes.com/sites/bernardmarr/2018/02/14/the-key-definitions-of-artificial-intelligence-ai-that-explain-its-importance/

Pan, J., Ding, S., Wu, D., Yang, S., & Yang, J. (2019). Exploring behavioural intentions toward smart healthcare services among medical practitioners: A technology transfer perspective. *International Journal of Production Research, 57*(18), 5801–5820.

Paranjape, K., Schinkel, M., NannanPanday, R. N., Car, J., &Nanayakkara, P. (2019). Introducing artificial intelligence training in medical education. JMIR Medical Education, 5(2). https://doi, e16048. https://doi.org/10.2196/16048

Peres, Renana, Schreier, M., Schweidel, D., &Sorescu, A. (2023). On ChatGPT and beyond: How generative artificial intelligence may affect research, teaching, and practice. International Journal of Research in Marketing, 40(2), 269–275. https://doi.org/10.1016/j.ijresmar.2023.03.001

Priore, P., Ponte, B., Rosillo, R., & de la Fuente, D. (2019). Applying machine learning to the dynamic selection of replenishment policies in fast-changing supply chain environments. *International Journal of Production Research, 57*(11), 3663–3677.

Razzaghi, T., Safro, I., Ewing, J., Sadrfaridpour, E., & Scott, J. D. (2019). Predictive models for bariatric surgery risks with imbalanced medical datasets. Annals of Operations Research. *280*, 1–18.

Razack, H. I. A., Mathew, S. T., Saad, F. F. A., &Alqahtani, S. A. (2021). Artificial intelligence-assisted tools for redefining the communication landscape of the scholarly world. Science Editing, 8(2), 134-144. https://doi.org/10.6087/kcse.244

Romanova, T., Stoyan, Y., Pankratov, A., Litvinchev, I., Avramov, K., Chernobryvko, M., et al. (2019). Optimal layout of ellipses and its applicationfor additive manufacturing. International Journal of Production Research. https://doi.org/10.1080/00207543.2019.1697836

Tariq Alqahtani , Hisham A. Badreldin , Mohammed Alrashed ,Abdulrahman I. Alshaya, Sahar S. Alghamdi , Khalid bin Saleh ,Shuroug A. Alowais , Omar A. Alshaya , Ishrat Rahman , Majed S. Al Yami ,&Abdulkareem M. Albekairy(2023).The emergent role of artificial intelligence, natural learning processing, andlarge language models in higher education and research. Research in Social and Administrative Pharmacy.19 ,1236–1242. https://doi.org/10.1016/j.sapharm.2023.05.016

Tennant, J. P., & Ross-Hellauer, T. (2020). The limitations to our understanding of peer review. Research Integrity and Peer Review, 5(1), 6. https://doi.org/10.1186/s41073-020-00092-1

Yassine, S., Esghir, M., &Ibrihich, O. (2023). Using artificial intelligence tools in judicial domain and the evaluation of their impact on the prediction of judgements. Procedia Computer Science, 220, 1021–1026. https://doi.org/10.1016/j.procs.2023.03.142

Yeung, J. A., Krajevic, Z., Balston, A., Idowu, E., Dobson, R., &Teo, J(2023). AI chatbots not yet ready for clinical use. medRxiv. https://doi.org/10.1101/2023.03.02.23286705

Yu-Cheng Wang , Tin-ChihToly Chen , Min-Chi Chiu , An improved explainable artificial intelligence tool in healthcare for hospital recommendation. https://doi.org/10.1016/j.health.2023.100147. Healthcare Analytics,Volume 3, November 2023, 100147

Solangi YA, Solangi ZA, Aarain S, Abro A, Mallah GA, Shah A(2019). Review on naturalLanguage Processing (NLP) and its toolkits for opinion mining and sentimentanalysis. In: *2018* IEEE 5th International Conference on Engineering Technologies and Applied Sciences. ICETAS; 2018. https://doi.org/10.1109/ICETAS.2018.8629198